Introduction to Data Science and AI
Assignment 1

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Work distribution

Georgi - Initial data fetching and initial plots(a.), b., c., report (b., c.)
Ilia – Refactoring the code, d., e., report (a., d., e.)
GitHub repo: https://github.com/Ilia-tod29/Introduction_to_Data_Science_and_AI

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Scatter plot of GDP per capita vs life expectancy (a.)

We used three datasets from Our World in Data. Our program downloads them directly as CSV files, so anyone can re-run it and get the same data:

- Life expectancy at birth; indicator: life_expectancy_0
- GDP per capita (World Bank); indicator: ny_gdp_pcap_pp_kd
- GDP (World Bank), for d.; indicator: ny_gdp_mktp_pp_kd

We have chosen the World Bank GDP series measured in PPP-adjusted constant 2021 dollars because it accounts for both inflation and differences in price levels between countries.

Combining the data

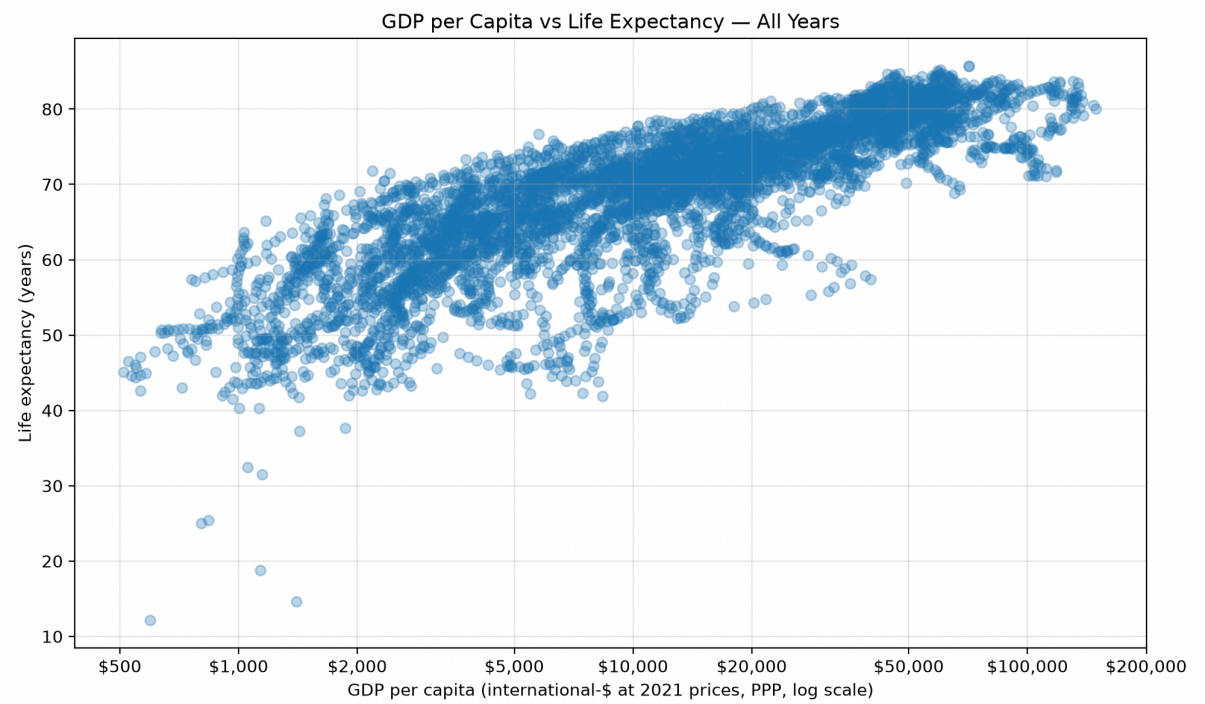
- Join key: We joined the datasets on country code + year, not on country name. ISO codes are stable, while names can be spelled differently in different files.
- Inner join between life expectancy and GDP per capita: As a result, the plotted period is 1990–2023. Life expectancy goes back to 1543, but GDP per capita starts in 1990, and life expectancy stops in 2023.
- Left join for total GDP, so that we add the total GDP to our combined dataset.

Data cleaning

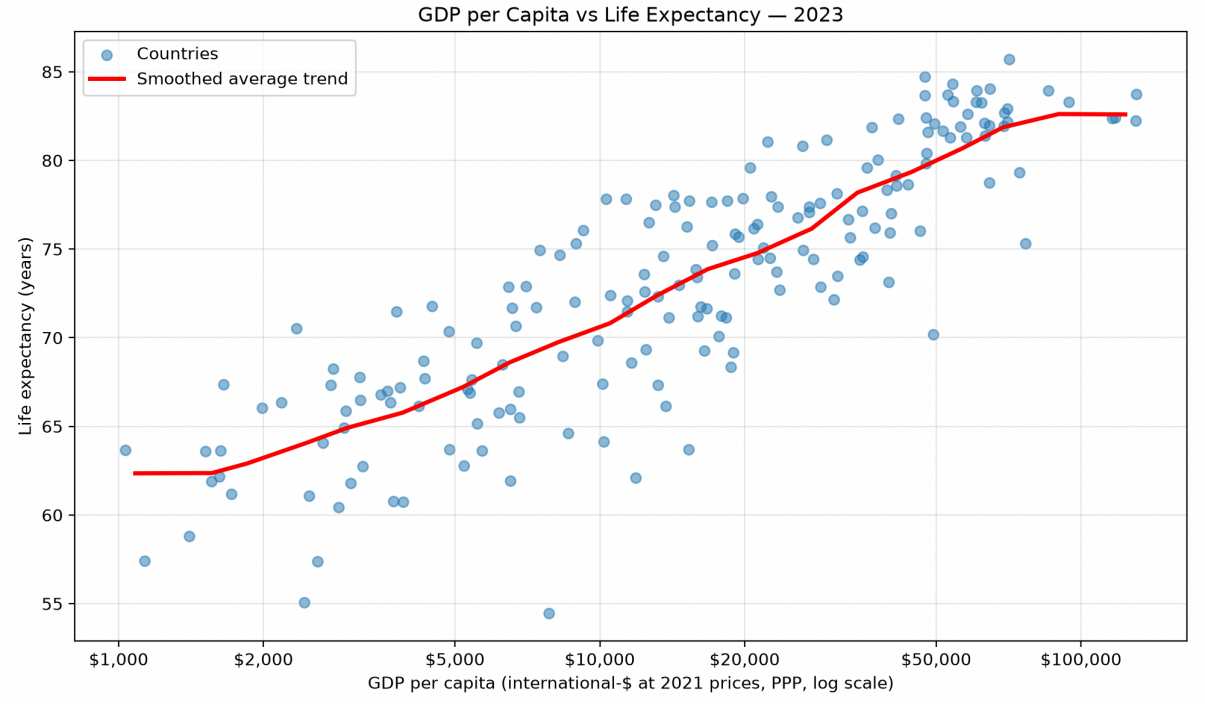
- Rows with missing or non-positive values removed.
- Aggregates removed. OWID marks regions and groups with codes starting with OWID_
- Kosovo kept as an exception. Kosovo has no ISO code, so OWID gives it OWID_KOS, but it is a country, not an aggregate.
- Dependent territories removed. They have their own ISO codes, so the aggregate filter keeps them, but they belong to another state. Small rich territories would otherwise crowd the top of the GDP per capita ranking and distort questions b–e. Kosovo and Palestine are kept as partially recognized states.

The figures

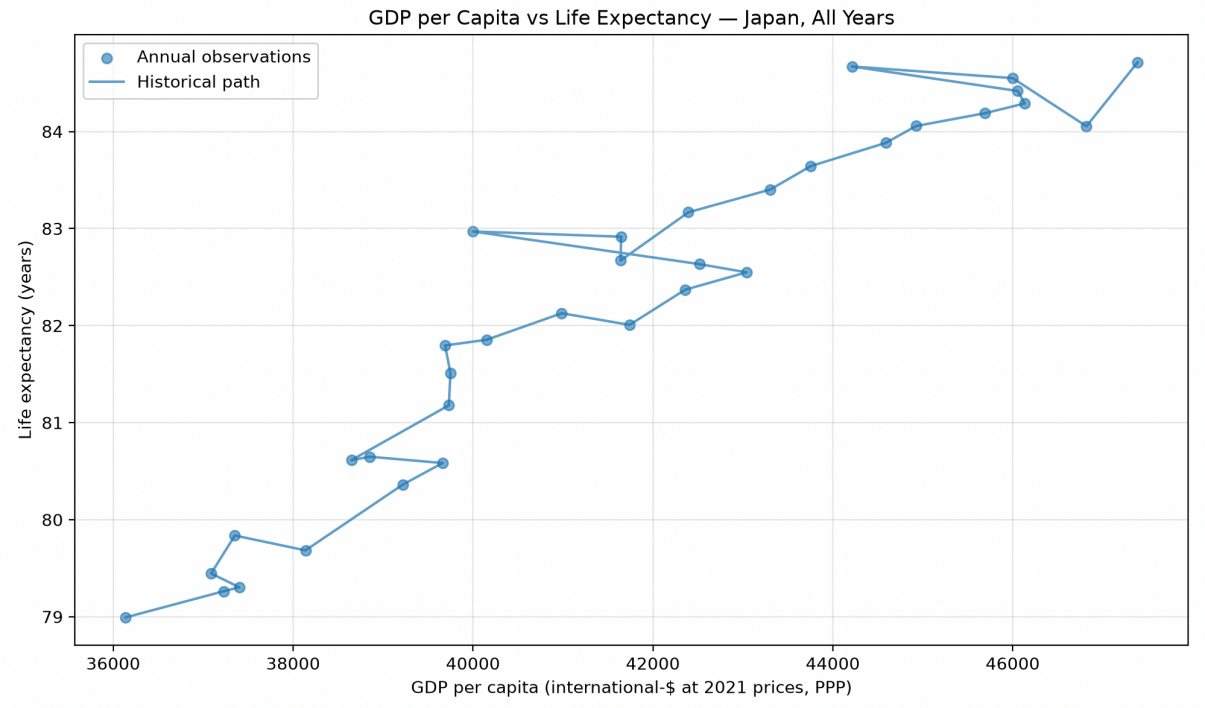
- Figure 1, all countries, 1990–2023. Each point is one country in one year. It shows the general pattern, but points from the same country are strongly correlated, and different decades get mixed together.



- Figure 2, all countries in 2023 (main figure). Each point is one country, all measured in the same year, so the countries are compared fairly. The red line is a smoothed average trend: We split GDP per capita into 20 equal-width bins on the log scale, averaged each bin, and applied a 5-bin rolling mean.



- Figure 3, Japan, 1990–2023. Looking at one country removes differences between countries (geography, climate, health system) and shows how income and life expectancy changed together over time. The points are joined in year order to show the path over time.



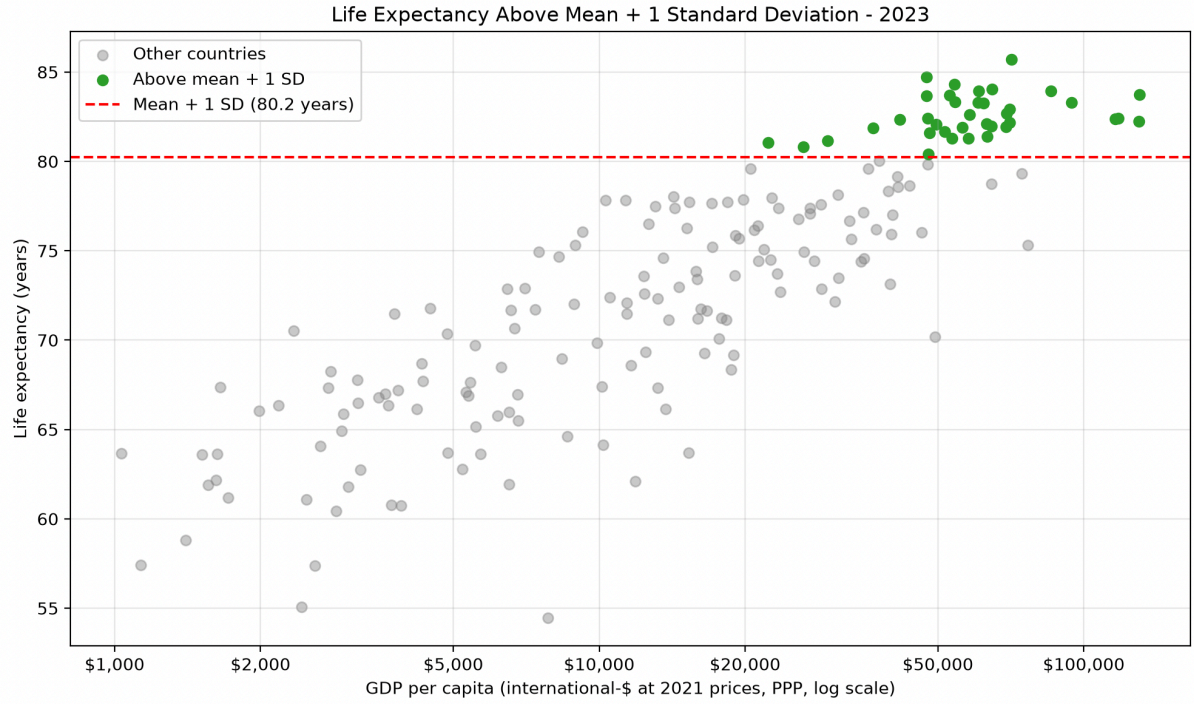
Assumptions

- OWID/World Bank figures are accurate and comparable between countries.
- PPP adjustment properly corrects for differences in price levels.
- Life expectancy at birth is a reasonable summary of a population's health.
- The relationship shown is a correlation, not causation. Things like education, healthcare spending and inequality affect both variables.

Observations

Life expectancy rises steeply with GDP per capita at low incomes and levels off at high incomes, at roughly 80–85 years above about $40,000. That is the classic "Preston curve". On the log scale, the relationship looks roughly linear, which supports using that scale.

Question b. Which countries have a life expectancy higher than one standard deviation above the mean?



Mean: 73.1 years, standard deviation: 7.1 years Cutoff (mean + 1 SD): 80.2 years 37 of 186 countries are above the cutoff. The first 6 are listed below, and the rest can be seen in the script output:

Country	**Life expectancy (years)**	**GDP per capita (int-$, 2021 prices)**
San Marino	85.7	$70,952
Japan	84.7	$47,387
South Korea	84.3	$54,029
Andorra	84.0	$64,631
Switzerland	84.0	$85,383
Australia	83.9	$60,685
…	…	…

Explanation

I used 2023 so that all countries are compared at the same point in time. The mean life expectancy across 186 countries is 73.1 years, with a standard deviation of 7.1 years, so the cutoff is 80.2 years.

The 37 countries above the cutoff make up about 20% of all countries. That is slightly higher than the roughly 16% expected from a bell-shaped distribution, because a few countries with low life expectancy, down to 54.5 years, pull the average down.

The table shows the top six, and the graph highlights all 37 in green above the red dashed line.

All 37 countries are above the median GDP per capita ($16,102), which fits what we saw in Graph 2.

Question B's "cutoff" is defined by the question itself, while Question C's "cutoff" uses the same threshold as d and e.

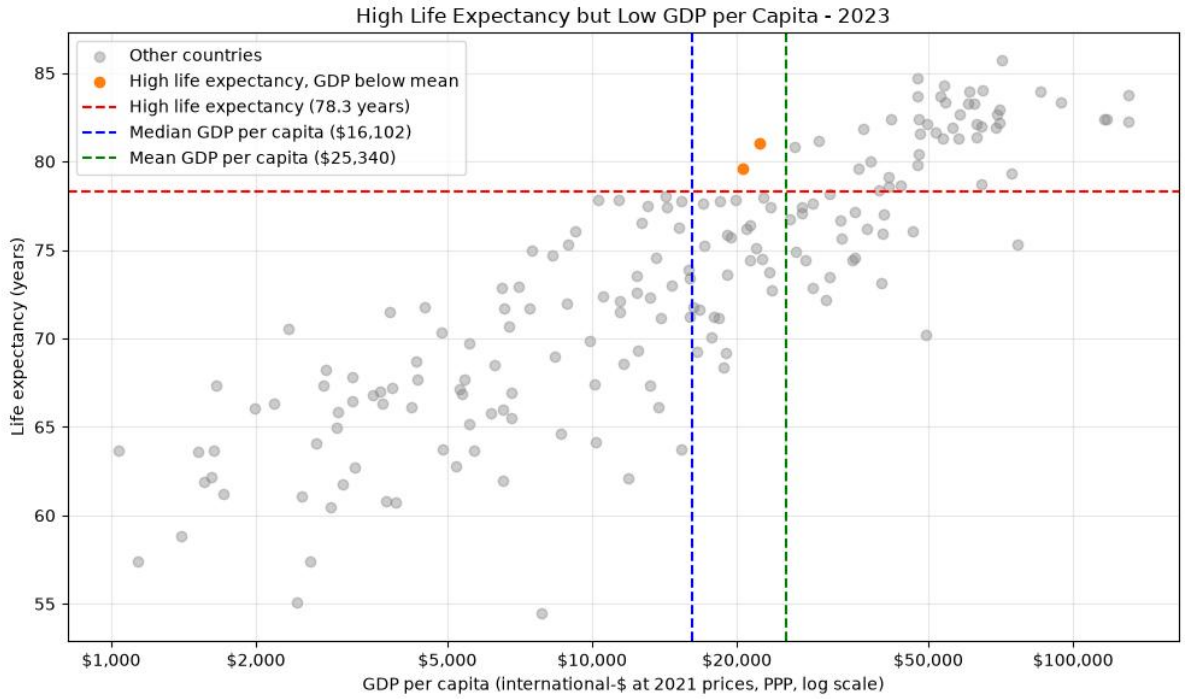
Countries are counted equally regardless of population, so Maldives counts the same as China. Small states like San Marino and Andorra also appear in the lists.

Limitations for question b

39 places with a 2023 life expectancy value were excluded because the World Bank has no GDP data for them. Most are dependent territories (e.g. Reunion, Guam). The rest are Monaco, Liechtenstein, Taiwan, Cuba, Venezuela and a few others.

Monaco, Liechtenstein and Taiwan would very likely have been above the cutoff, so the list of 37 is a slight undercount.

Question C. Which countries have high life expectancy but have low GDP? Motivate how you have chosen to define “high” and “low”.



High life expectancy: at least 78.3 years (upper quartile of 2023, same threshold as in questions d and e). Low GDP per capita, strict: below the median, $16,102. Low GDP per capita, relaxed: below the mean, $25,340.

Strict definition: 0 countries. Relaxed definition: 2 countries.

Country	**Life expectancy (years)**	**GDP per capita (int-$, 2021 prices)**
Maldives	81.0	$22,333
Albania	79.6	$20,502

Explanation

I used 2023 so that all countries are compared at the same point in time. "High" life expectancy means the upper quartile, at least 78.3 years, so the whole report uses one definition of "high".

For "low" GDP per capita, I first used the median ($16,102), meaning the poorer half of countries. No country meets both conditions: the poorest country with high life expectancy is Albania, at $20,502, which is still above the median. In 2023, no country is in the poorer half and also in the top quartile for life expectancy.

To find the closest exceptions, I relaxed "low" to below the mean ($25,340). The mean is higher than the median because a few very rich countries pull it up. Two countries pass: Maldives and Albania.

This fits Graph 2: income and life expectancy rise together, so high life expectancy at a low income is very rare. Costa Rica ($26,379) just misses the relaxed cutoff, which shows the result depends on where the line is drawn.

Limitations for question c

The same 39 places were excluded because the World Bank has no GDP data for them. The result of question c would not change if they were included:

- Monaco, Liechtenstein and Taiwan are high-income countries (general knowledge, not measured in our data), so they would not count as "low GDP".
- Cuba, at 78.1 years, is just below the 78.3 threshold.
- The other excluded countries with low life expectancy, such as Venezuela, are far below it

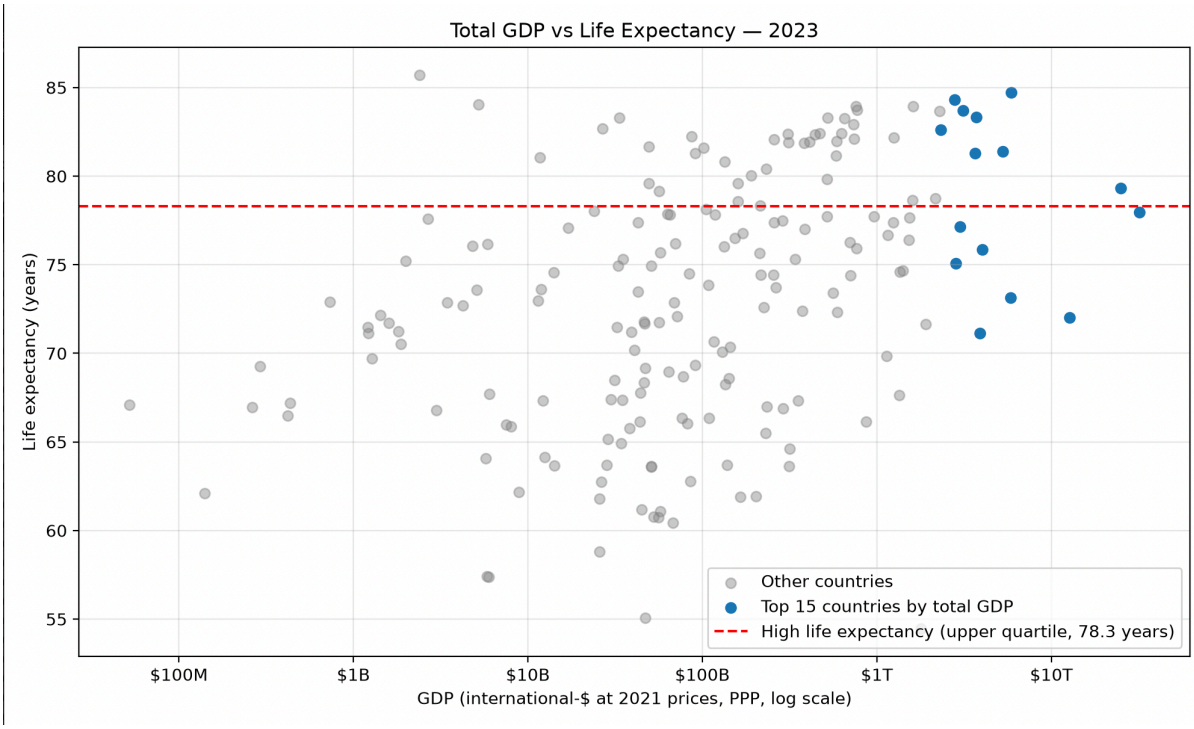
Does every strong economy (normally indicated by GDP) have high life expectancy? (d.)

Definitions/Assumptions

A "strong economy" is one of the 15 largest economies by total GDP (PPP, 2023). "High" life expectancy means the upper quartile of countries in 2023, which is ≥ 78.3 years. We use the latest year so that countries are compared at the same point in time.

Answer

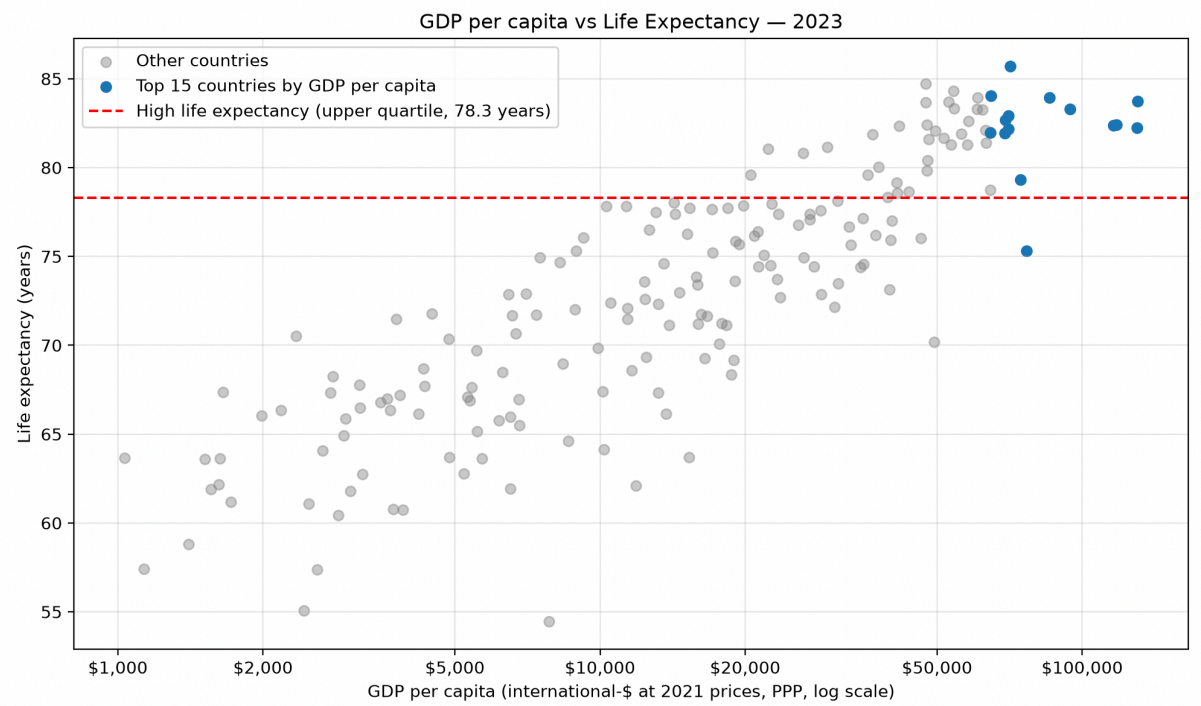
No. Only 8 of the top 15 are above the threshold: the US, Japan, Germany, France, the UK, Italy, South Korea and Canada. The other 7 are below it.
Total GDP measures how big an economy is, and that depends heavily on population. China and India rank high mainly because they have so many people, not because each person is wealthy.



Related to question d, what would happen if you use GDP per capita as an indicator of a strong economy? (e.)

Answer

With the same threshold, 14 of the top 15 have high life expectancy. There is just one exception – Brunei, which is not annotated in the figure, but it is in the data itself.



Insight from comparing d and e

Life expectancy depends on how much wealth there is per person, not on the total size of the economy. GDP per capita says much more about living standards, such as healthcare, nutrition and sanitation, so it predicts life expectancy far better. The top-15 list also changes almost completely: only the US is in both. It switches from large-population countries to small, rich countries like Singapore, Luxembourg and Qatar. Even so, a high GDP per capita doesn't guarantee high life expectancy. Brunei, and the US being only just above the line (79.3), suggest that inequality and how wealth is spent also matter.
From the code result (print after the plots):
Top 15 countries by total GDP in 2023:

- Country GDP Life expectancy
- China 32005133203227 77.9529
- United States 25038301109944 79.3043
- India 12756468651839 72.0026
- Japan 5900497545321 84.7123
- Russia 5834620196473 73.1541
- Germany 5258774650739 81.3777
- Brazil 4028525459920 75.8482
- Indonesia 3905715830893 71.1457
- France 3712391079956 83.3253
- United Kingdom 3660893459405 81.3015
- Italy 3116881036053 83.7157
- Turkey 2992310786629 77.1557

- Mexico 2836749919768 75.0690
- South Korea 2793998855187 84.3288
- Canada 2326386554601 82.6302

Top 15 countries by GDP per capita in 2023:

- Country GDP per capita Life expectancy
- Singapore 130296.840 83.7362
- Luxembourg 130048.920 82.2294
- Ireland 117659.730 82.4118
- Qatar 116159.140 82.3680
- Norway 94364.960 83.3078
- Switzerland 85383.360 83.9536
- Brunei 76616.700 75.3266
- United States 74351.670 79.3043
- San Marino 70951.790 85.7062
- United Arab Emirates 70239.836 82.9091
- Netherlands 70198.950 82.1576
- Iceland 69268.125 82.6911
- Denmark 68993.910 81.9328
- Andorra 64631.297 84.0406
- Austria 64536.207 81.9559

Appendix: Python code (main.py)

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# ============================================================
# Assignment 1
# By Ilia Todorov FN: 2MI3400906 & Georgi Karamishev FN: 2MI3200988
# ============================================================

# ============================================================
# 1. DOWNLOAD AND PREPARE THE DATA
# ============================================================

# Dataset URLs:
# https://ourworldindata.org/grapher/life-expectancy
# https://ourworldindata.org/grapher/gdp-per-capita-worldbank
# https://ourworldindata.org/grapher/gdp-worldbank
```

```
life_url = (
    "https://ourworldindata.org/grapher/life-expectancy.csv"
    "?v=1&csvType=full&useColumnShortNames=true"
)

gdp_url = (
    "https://ourworldindata.org/grapher/gdp-per-capita-worldbank.csv"
    "?v=1&csvType=full&useColumnShortNames=true"
)

# Total GDP is needed for question d. It comes from the same World Bank
# source and uses the same unit (international-$ at 2021 prices, PPP)
# as GDP per capita, so the two measures can be compared directly.
gdp_total_url = (
    "https://ourworldindata.org/grapher/gdp-worldbank.csv"
    "?v=1&csvType=full&useColumnShortNames=true"
)

user_agent = {"User-Agent": "Our World In Data data fetch/1.0"}

life_df = pd.read_csv(life_url, storage_options=user_agent)
gdp_df = pd.read_csv(gdp_url, storage_options=user_agent)
gdp_total_df = pd.read_csv(gdp_total_url, storage_options=user_agent)

# Rename columns to make them easier to understand.
life_df = life_df.rename(columns={
    "entity": "Country",
    "code": "Code",
    "year": "Year",
    "life_expectancy_0": "Life expectancy"
})

gdp_df = gdp_df.rename(columns={
    "entity": "Country",
    "code": "Code",
    "year": "Year",
    "ny_gdp_pcap_pp_kd": "GDP per capita"
})

gdp_total_df = gdp_total_df.rename(columns={
    "code": "Code",
    "year": "Year",
    "ny_gdp_mktp_pp_kd": "GDP"
})

# Keep only the columns needed for the analysis.
life_df = life_df[
    ["Country", "Code", "Year", "Life expectancy"]
```

```
]

gdp_df = gdp_df[
    ["Country", "Code", "Year", "GDP per capita"]
]

gdp_total_df = gdp_total_df[
    ["Code", "Year", "GDP"]
]

# Combine GDP and life expectancy using country code and year.
# An inner join keeps only observations present in both datasets.
df = pd.merge(
    life_df,
    gdp_df,
    on=["Code", "Year"],
    how="inner",
    suffixes=("_life", "_gdp")
)

# Add total GDP. A left join keeps every observation used in the
# scatter plots, even where total GDP is missing.
df = pd.merge(
    df,
    gdp_total_df,
    on=["Code", "Year"],
    how="left"
)

# Remove observations without valid GDP or life expectancy.
df = df.dropna(subset=["GDP per capita", "Life expectancy"])

# Keep positive values and exclude aggregate entities without country codes.
df = df[
    (df["GDP per capita"] > 0) &
    (df["Life expectancy"] > 0) &
    df["Code"].notna()
].copy()

# Exclude aggregates that Our World in Data gives an "OWID_" code,
# such as World and the income groups. Kosovo also uses an "OWID_"
# code because it has no ISO code, but it is a country, so keep it.
is_aggregate = (
    df["Code"].str.startswith("OWID_") &
    (df["Code"] != "OWID_KOS")
)
df = df[~is_aggregate].copy()

# Exclude territories that are not independent countries. They have
```

```
# their own ISO codes, so the filter above keeps them, but they are
# part of another country (e.g. Greenland belongs to Denmark and
# Macao to China). Small, rich territories such as Bermuda and the
# Cayman Islands would otherwise fill the top of the GDP per capita
# ranking. Kosovo and Palestine are kept as partially recognised states.
territory_codes = [
    "ABW",  # Aruba (Netherlands)
    "BMU",  # Bermuda (United Kingdom)
    "CUW",  # Curacao (Netherlands)
    "CYM",  # Cayman Islands (United Kingdom)
    "FRO",  # Faroe Islands (Denmark)
    "GRL",  # Greenland (Denmark)
    "HKG",  # Hong Kong (China)
    "MAC",  # Macao (China)
    "PRI",  # Puerto Rico (United States)
    "SXM",  # Sint Maarten (Netherlands)
    "TCA",  # Turks and Caicos Islands (United Kingdom)
    "VIR",  # United States Virgin Islands (United States)
]
df = df[~df["Code"].isin(territory_codes)].copy()

# Keep one country name column after merging.
if "Country_life" in df.columns:
    df["Country"] = df["Country_life"]


# ============================================================
# 2. GRAPH 1 — ALL COUNTRIES, ALL AVAILABLE YEARS
#    Each point represents a country in a particular year.
#    Historical and modern observations are included together.
# ============================================================

plt.figure(figsize=(10, 6))

plt.scatter(
    df["GDP per capita"],
    df["Life expectancy"],
    alpha=0.3
)

plt.title("GDP per Capita vs Life Expectancy — All Years")
plt.xlabel("GDP per capita (international-$ at 2021 prices, PPP, log scale)")
plt.xscale("log")

# Label the log axis in plain dollars instead of powers of 10.
gdp_ticks = [500, 1000, 2000, 5000, 10000, 20000, 50000, 100000, 200000]
plt.xticks(gdp_ticks, [f"${tick:,}" for tick in gdp_ticks])
plt.minorticks_off()
plt.ylabel("Life expectancy (years)")
```

```
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()


# ============================================================
# 3. GRAPH 2 — ALL COUNTRIES IN 2023
#    Each point represents one country in 2023.
#    This is a defensible choice because it compares countries
#    at the same point in time, rather than mixing historical
#    and modern life expectancy values.
#
#    The red curve shows a smoothed average trend across
#    GDP-per-capita intervals. It helps reveal the general
#    relationship without displaying every average as a dot.
# ============================================================

selected_year = 2023

year_df = df[df["Year"] == selected_year].copy()

plt.figure(figsize=(10, 6))

# Show the individual countries.
plt.scatter(
    year_df["GDP per capita"],
    year_df["Life expectancy"],
    alpha=0.5,
    label="Countries"
)

# Divide GDP per capita into 20 intervals of equal width on the
# log scale, so each interval covers the same width on the x-axis.
year_df["GDP group"] = pd.cut(
    np.log10(year_df["GDP per capita"]),
    bins=20
)

# Calculate average GDP and life expectancy in each interval.
average_df = year_df.groupby(
    "GDP group",
    observed=True
).agg({
    "GDP per capita": "mean",
    "Life expectancy": "mean"
}).dropna()

# Sort the averages by GDP before smoothing.
average_df = average_df.sort_values("GDP per capita")
```

```
# Smooth the trend using a centered rolling average.
average_df["Smoothed life expectancy"] = (
    average_df["Life expectancy"]
    .rolling(window=5, center=True, min_periods=1)
    .mean()
)

# Draw the smoothed trend without individual markers.
plt.plot(
    average_df["GDP per capita"],
    average_df["Smoothed life expectancy"],
    color="red",
    linewidth=2.5,
    label="Smoothed average trend"
)

plt.title(f"GDP per Capita vs Life Expectancy — {selected_year}")
plt.xlabel("GDP per capita (international-$ at 2021 prices, PPP, log scale)")
plt.xscale("log")

# Label the log axis in plain dollars instead of powers of 10.
gdp_ticks = [1000, 2000, 5000, 10000, 20000, 50000, 100000]
plt.xticks(gdp_ticks, [f"${tick:,}" for tick in gdp_ticks])
plt.minorticks_off()
plt.ylabel("Life expectancy (years)")
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()


# ============================================================
# 4. GRAPH 3 — JAPAN OVER TIME
#    Using one country reduces geographical differences,
#    such as climate and location, allowing us to examine
#    how GDP per capita and life expectancy change together
#    within the same country over time.
# ============================================================

selected_country = "Japan"

country_df = df[
    df["Country"] == selected_country
].copy()

# Sort chronologically so the line follows Japan's history.
country_df = country_df.sort_values("Year")
```

```
plt.figure(figsize=(10, 6))

# Each point represents Japan in one year.
plt.scatter(
    country_df["GDP per capita"],
    country_df["Life expectancy"],
    alpha=0.6,
    label="Annual observations"
)

# Connect observations chronologically to show historical development.
plt.plot(
    country_df["GDP per capita"],
    country_df["Life expectancy"],
    linewidth=1.5,
    alpha=0.7,
    label="Historical path"
)

plt.title(
    f"GDP per Capita vs Life Expectancy — {selected_country}, All Years"
)
plt.xlabel("GDP per capita (international-$ at 2021 prices, PPP)")
plt.ylabel("Life expectancy (years)")
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()


# ============================================================
# 5. GRAPHS 4 AND 5 — QUESTIONS D AND E: STRONG ECONOMIES VS
#    LIFE EXPECTANCY IN 2023
#    The same plot is drawn twice: once with total GDP (question d)
#    and once with GDP per capita (question e) as the indicator of
#    a strong economy. The top countries by that indicator are
#    highlighted so we can see whether all of them are above the
#    "high life expectancy" line.
#
#    We consider "High" life expectancy to be the upper quartile: life
#    expectancy at least as high as 75% of countries in 2023. The same
#    threshold is used in both plots so the results can be compared.
# ============================================================

top_n = 15

high_life_threshold = year_df["Life expectancy"].quantile(0.75)


def plot_strong_economies(column, title_name, xlabel, ticks, tick_labels):
```

```
    """Plot an economic indicator against life expectancy for one year.

    column      -- the indicator column, e.g. "GDP" or "GDP per capita"
    title_name  -- how the indicator is named in the title and legend
    xlabel      -- x-axis label, including the unit
    ticks       -- positions of the x-axis ticks
    tick_labels -- text shown at each x-axis tick
    """
    indicator_df = year_df.dropna(subset=[column]).copy()

    top_df = indicator_df.nlargest(top_n, column)

    plt.figure(figsize=(10, 6))

    # Show all countries in grey for context.
    plt.scatter(
        indicator_df[column],
        indicator_df["Life expectancy"],
        color="grey",
        alpha=0.4,
        label="Other countries"
    )

    # Highlight the top countries by the chosen indicator.
    plt.scatter(
        top_df[column],
        top_df["Life expectancy"],
        color="tab:blue",
        label=f"Top {top_n} countries by {title_name}"
    )

    # Mark the threshold for "high" life expectancy.
    plt.axhline(
        high_life_threshold,
        color="red",
        linestyle="--",
        label=f"High life expectancy (upper quartile, {high_life_threshold:.1f}
years)"
    )

    # Capitalise only the first letter so "GDP" stays in capitals.
    plt.title(
        f"{title_name[0].upper()}{title_name[1:]} vs Life Expectancy — "
        f"{selected_year}"
    )
    plt.xlabel(xlabel)
    plt.xscale("log")
    plt.xticks(ticks, tick_labels)
    plt.minorticks_off()
```

```
    plt.ylabel("Life expectancy (years)")
    plt.grid(True, alpha=0.3)
    plt.legend()
    plt.tight_layout()
    plt.show()

    print(f"Top {top_n} countries by {title_name} in {selected_year}:")
    print(top_df[["Country", column, "Life expectancy"]].to_string(index=False))
    print()


# Question d: total GDP, with the log axis in billions and trillions.
plot_strong_economies(
    column="GDP",
    title_name="total GDP",
    xlabel="GDP (international-$ at 2021 prices, PPP, log scale)",
    ticks=[1e8, 1e9, 1e10, 1e11, 1e12, 1e13],
    tick_labels=["$100M", "$1B", "$10B", "$100B", "$1T", "$10T"]
)

# Question e: GDP per capita, with the same ticks as graph 2.
plot_strong_economies(
    column="GDP per capita",
    title_name="GDP per capita",
    xlabel="GDP per capita (international-$ at 2021 prices, PPP, log scale)",
    ticks=gdp_ticks,
    tick_labels=[f"${tick:,}" for tick in gdp_ticks]
)
# ============================================================
# 6. QUESTION B - LIFE EXPECTANCY MORE THAN ONE STANDARD
#    DEVIATION ABOVE THE MEAN (2023)
# ============================================================

mean_life = year_df["Life expectancy"].mean()
std_life = year_df["Life expectancy"].std()
cutoff_b = mean_life + std_life

above_df = year_df[year_df["Life expectancy"] > cutoff_b].sort_values(
    "Life expectancy", ascending=False
)

print(f"Question b - {selected_year}")
print(f"Mean: {mean_life:.1f} years, standard deviation: {std_life:.1f} years")
print(f"Cutoff (mean + 1 SD): {cutoff_b:.1f} years")
print(f"{len(above_df)} of {len(year_df)} countries are above the cutoff:")
print(above_df[["Country", "Life expectancy", "GDP per capita"]].round({"Life
expectancy": 1, "GDP per capita": 0}).to_string(index=False))
print()
```

```
plt.figure(figsize=(10, 6))
plt.scatter(
    year_df["GDP per capita"], year_df["Life expectancy"],
    color="grey", alpha=0.4, label="Other countries"
)
plt.scatter(
    above_df["GDP per capita"], above_df["Life expectancy"],
    color="tab:green", label="Above mean + 1 SD"
)
plt.axhline(
    cutoff_b, color="red", linestyle="--",
    label=f"Mean + 1 SD ({cutoff_b:.1f} years)"
)
plt.title(f"Life Expectancy Above Mean + 1 Standard Deviation - {selected_year}")
plt.xlabel("GDP per capita (international-$ at 2021 prices, PPP, log scale)")
plt.xscale("log")
plt.xticks(gdp_ticks, [f"${tick:,}" for tick in gdp_ticks])
plt.minorticks_off()
plt.ylabel("Life expectancy (years)")
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()


# ============================================================
# 7. QUESTION C - HIGH LIFE EXPECTANCY BUT LOW GDP PER CAPITA (2023)
#    High: upper quartile of life expectancy (same as d and e).
#    Low (strict): below the median GDP per capita.
#    Low (relaxed): below the mean GDP per capita.
# ============================================================

median_gdp = year_df["GDP per capita"].median()
mean_gdp = year_df["GDP per capita"].mean()
high_df = year_df[year_df["Life expectancy"] >= high_life_threshold]

strict_df = high_df[high_df["GDP per capita"] < median_gdp]
relaxed_df = high_df[high_df["GDP per capita"] < mean_gdp].sort_values(
    "Life expectancy", ascending=False
)

print(f"Question c - {selected_year}")
print(f"High life expectancy: at least {high_life_threshold:.1f} years")
print(f"Strict low GDP (below median ${median_gdp:,.0f}): {len(strict_df)}
countries")
print(f"Relaxed low GDP (below mean ${mean_gdp:,.0f}): {len(relaxed_df)}
countries")
print(relaxed_df[["Country", "Life expectancy", "GDP per capita"]]
      .round({"Life expectancy": 1, "GDP per capita": 0})
```

```
      .to_string(index=False))
print()

plt.figure(figsize=(10, 6))
plt.scatter(year_df["GDP per capita"], year_df["Life expectancy"],
            color="grey", alpha=0.4, label="Other countries")
plt.scatter(relaxed_df["GDP per capita"], relaxed_df["Life expectancy"],
            color="tab:orange", label="High life expectancy, GDP below mean")
plt.axhline(high_life_threshold, color="red", linestyle="--",
            label=f"High life expectancy ({high_life_threshold:.1f} years)")
plt.axvline(median_gdp, color="blue", linestyle="--",
            label=f"Median GDP per capita (${median_gdp:,.0f})")
plt.axvline(mean_gdp, color="green", linestyle="--",
            label=f"Mean GDP per capita (${mean_gdp:,.0f})")
plt.title(f"High Life Expectancy but Low GDP per Capita - {selected_year}")
plt.xlabel("GDP per capita (international-$ at 2021 prices, PPP, log scale)")
plt.xscale("log")
plt.xticks(gdp_ticks, [f"${tick:,}" for tick in gdp_ticks])
plt.minorticks_off()
plt.ylabel("Life expectancy (years)")
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()
```